Ordinary Differential Equations Note

Differential Equations with Boundary-Value Problems, 9/e (Metric Edition)

4 Higher-Order Differential Equations

Higher-order(高階) means the order of the DE is $n \geq 2$.

4.1 Preliminary Theory - Linear Equations

4.1.1 Initial-Value and Boundary-Value Problems

Definition. An nth-order **initial-value problem** (IVP) is

$$\begin{aligned} \text{Solve: } & a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x) \\ \text{Subject to: } & y(x_0) = y_0,\ y'(x_0) = y_1,\ \ldots,\ y^{(n-1)}(x_0) = y_{n-1} \end{aligned} \tag{1}$$

with n initial conditions(初始條件, I.C.).

Remark.
1. $a_n(x)$, $a_{n-1}(x)$, $\ldots$, $a_0(x)$ are the coefficients(係數) of the equation.
2. All the n initial conditions are given at the **same** point x_0.
3. $g(x) = 0$ or $g(x) \neq 0$. (See 4.1.2 and 4.1.3)

Example. Consider

$$y'' + 2y' + 3y = \sin x.$$

It is a 2nd-order linear DE with

$$a_2(x) = 1, \quad a_1(x) = 2, \quad a_0(x) = 3, \quad g(x) = \sin x \neq 0.$$

Since $g(x) \neq 0$, the equation is **nonhomogeneous**(非齊次).

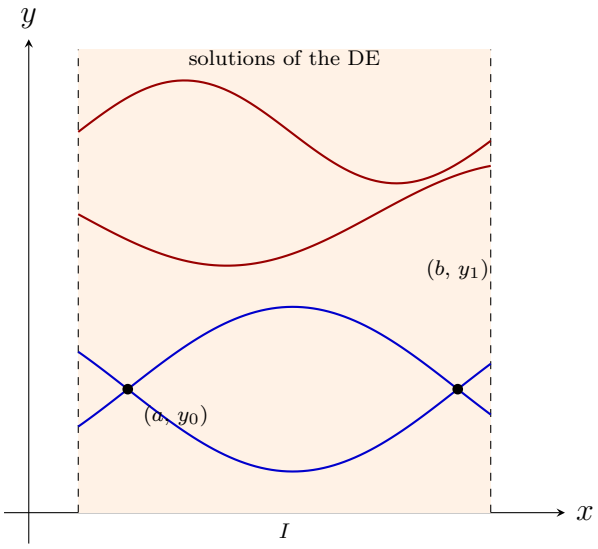
Theorem (Existence of a Unique Solution)**.** Let $a_n(x)$, $a_{n-1}(x)$, $\ldots$, $a_0(x)$ and $g(x)$ be continuous on an interval I, and $a_n(x) \neq 0$ for all x on I. If $x = x_0$ is any point in this interval, then a solution $y(x)$ of the IVP (1) exists on the interval and is unique.

Definition. A problem of the form

$$\begin{aligned} \text{Solve: } & a_2(x)\frac{d^2 y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x) \\ \text{Subject to: } & y(a) = y_0,\ y(b) = y_1 \end{aligned}$$

is called a **boundary-value problem** (BVP, 邊界值問題).

Remark. The difference between IVP and BVP: in an IVP, all conditions are given at the same point x_0; in a BVP, the conditions are given at two **different** points a and b (the boundary of the interval I).
So a solution of the BVP is a solution curve passing through the two points $(a,\ y_0)$ and $(b,\ y_1)$:



4.1.2 Homogeneous Equations

Definition. The nth-order linear DE

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = 0$$

(i.e. $g(x) = 0$) is said to be **homogeneous**(齊次).

Example. $y'' + 3xy' + x^2y = 0$ is a homogeneous equation.

Theorem (Superposition Principles(疊加原理) - Homogeneous Equations)**.** Let $y_1,\ y_2,\ \ldots,\ y_k$ be solutions of the homogeneous nth-order differential equation on an interval I. Then the linear combination

$$y = c_1y_1(x) + c_2y_2(x) + \cdots + c_ky_k(x),$$

where $c_i,\ i = 1,\ 2,\ \ldots,\ k$ are arbitrary constants, is also a solution on the interval.

Corollary. (a) $y = cy_1$ is also a solution if y_1 is a solution.

(b) A homogeneous linear DE always possesses the **trivial solution** $y = 0$.

Example. For $y'' + 3xy' + x^2y = 0$, if y_1 and y_2 are solutions and they are **linearly independent**(線性獨立), then

$$y_h = c_1y_1 + c_2y_2$$

is the homogeneous solution(齊次解).
Similarly, for the 3rd-order equation

$$y''' + 2xy'' + 5y' + 6x^2y = 0,$$

if y_1, y_2, y_3 are linearly independent solutions, then

$$y_h = c_1y_1 + c_2y_2 + c_3y_3.$$

Question: How do we test whether y_1, y_2, y_3 are independent(獨立) or dependent(相依)?

Definition. Recall from linear algebra: a set of functions $f_1(x)$, $f_2(x)$, ..., $f_n(x)$ is **linearly dependent**(線性相依) on I if there exist constants c_1, c_2, ..., c_n, not all zero, such that

$$c_1f_1(x) + c_2f_2(x) + \cdots + c_nf_n(x) = 0$$

for every x in I. Otherwise, the set is **linearly independent**(線性獨立).

Definition. The **Wronskian** of y_1, y_2, y_3 is the determinant(行列式)

$$W(y_1, y_2, y_3) = \det\begin{pmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{pmatrix} = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}.$$

In general, for n functions,

$$W(y_1, y_2, \ldots, y_n) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}.$$

Theorem. If $W(y_1, y_2, y_3) \neq 0$ for all x in I, then y_1, y_2, y_3 are **linearly independent**.

Definition. Any set y_1, y_2, ..., y_n of n linearly independent solutions of the homogeneous nth-order linear DE on an interval I is said to be a **fundamental set of solutions**(基本解集合) on the interval.

4.1.3 Nonhomogeneous Equations

Definition. Any function y_p, free of arbitrary parameters, that satisfies the nonhomogeneous(非齊次) equation is called a **particular solution**(特解).

Example. $y_p(x)$ is a particular solution of $y'' + 2y' + 3y = x^2$ if

$$y_p'' + 2y_p' + 3y_p = x^2.$$

To solve

$$y'' + a_1(x)y' + a_2(x)y = f(x),$$

we will use the following steps:

(i) Find $y_h(x)$: the **homogeneous solution**, also called the **complementary solution**(補解), which satisfies

$$y_h'' + a_1(x)y_h' + a_2(x)y_h = 0.$$

(ii) Find $y_p(x)$: the **particular solution**, which satisfies

$$y_p'' + a_1(x)y_p' + a_2(x)y_p = f(x).$$

The general solution is

$$y(x) = y_h(x) + y_p(x).$$

Remark. It is the same idea as in Section 2.3: y_h is the output due to the initial condition (内部), and y_p is the output due to the external input $f(x)$ (外部).

Example. Solve $y'' - 9y = 0$.

(i) Find the fundamental set of solutions y_1 and y_2.

(ii) Test independency.

(iii) If independent, then $y_h = c_1y_1 + c_2y_2$.

Solution. Step 1. Since

$$y'' - 9y = 0 \implies y'' = 9y,$$

we need a function whose second derivative is 9 times itself.
Guess $y = e^{3x}$, then

$$\left(e^{3x}\right)' = 3e^{3x}, \quad \left(e^{3x}\right)'' = 3^2e^{3x} = 9e^{3x}.$$

Thus $y_1 = e^{3x}$ is a solution.
Guess $y = e^{-3x}$, then

$$\left(e^{-3x}\right)' = (-3)e^{-3x}, \quad \left(e^{-3x}\right)'' = (-3)(-3)e^{-3x} = 9e^{-3x}.$$

Thus $y_2 = e^{-3x}$ is also a solution.

Step 2. Test independence:

$$\begin{aligned} W(y_1,\, y_2) &= \begin{vmatrix} e^{3x} & e^{-3x} \\ 3e^{3x} & -3e^{-3x} \end{vmatrix} \\ &= e^{3x} \cdot \left(-3e^{-3x}\right) - e^{-3x} \cdot 3e^{3x} \\ &= -3 - 3 = -6 \neq 0. \end{aligned}$$

Therefore y_1, y_2 are linearly independent, and $\{e^{3x},\, e^{-3x}\}$ is a fundamental set of solutions.

Step 3. Hence

$$y_h = c_1y_1 + c_2y_2 = c_1e^{3x} + c_2e^{-3x}.$$

Since the equation is homogeneous, $y = c_1e^{3x} + c_2e^{-3x}$ is the general solution for constants c_1, c_2.

4.2 Reduction of Order

Reduction of order(降階法) is a method to find **another solution**(求另一個解). Suppose y_1 is a known solution of a homogeneous 2nd-order linear DE. We let

$$y_2 = y_1 \cdot u(x),$$

where $u(x)$ is a function(函數) to be determined, and y_2 is another solution.

Example. Find the fundamental set of solutions of

$$\left(x^2 - x\right) y'' - xy' + y = 0,$$

where $y_1 = x$ is a solution.

Sol. Let $y_2 = y_1 \cdot u(x) = x \cdot u(x)$ be another solution, then

$$\left(x^2 - x\right) y_2'' - xy_2' + y_2 = 0.$$

By product rule,

$$\begin{cases} y_2' = (xu)' = x'u + xu' = u'x + u \\ y_2'' = (xu)'' = (u'x + u)' = u''x + 2u' \end{cases}$$

Substituting y_2, y_2', y_2'' into the equation, we obtain

$$\begin{aligned} & \left(x^2 - x\right)(u''x + 2u') - x\left(u'x + u\right) + xu = 0 \\ \Longrightarrow\ & x\left(x^2 - x\right)u'' + \left(2x^2 - 2x - x^2\right)u' - xu + xu = 0 \\ \Longrightarrow\ & x\left(x^2 - x\right)u'' + \left(x^2 - 2x\right)u' = 0 \\ \Longrightarrow\ & \left(x^2 - x\right)u'' + (x - 2)u' = 0. \end{aligned}$$

The equation has no u term, so we can reduce the order(降階): let $v = u'$ (then $u = \int v\,dx$), and

$$\begin{aligned} \left(x^2 - x\right)v' + (x - 2)v = 0 \ &\Longrightarrow\ \left(x^2 - x\right)\frac{dv}{dx} = -(x - 2)v \\ &\Longrightarrow\ \frac{1}{v}dv = -\frac{x - 2}{x^2 - x}dx, \end{aligned}$$

which is a separable equation (Section 2.2). Then

$$\int \frac{1}{v}dv = -\int \frac{x - 2}{x^2 - x}dx + C.$$

By partial fractions,

$$\frac{x - 2}{x^2 - x} = \frac{x - 2}{x(x - 1)} = \frac{2}{x} + \frac{-1}{x - 1}.$$

Thus

$$\begin{aligned} \ln|v| &= -\int \frac{2}{x} - \frac{1}{x - 1}dx + C \\ &= -2\ln|x| + \ln|x - 1| + C \\ &= \ln\left|\frac{x - 1}{x^2}\right| + C \\ \Longrightarrow\ v &= e^C \cdot \frac{x - 1}{x^2}. \end{aligned}$$

We only need one function, so take $e^C = 1$:

(i) $v(x) = \dfrac{x-1}{x^2} = \dfrac{1}{x} - \dfrac{1}{x^2}.$

(ii) $u(x) = \displaystyle\int v(x)\,dx = \int \frac{1}{x} - \frac{1}{x^2}dx = \ln|x| + \frac{1}{x}.$

(iii) $y_2 = y_1 \cdot u(x) = x\left(\ln|x| + \dfrac{1}{x}\right) = x\ln|x| + 1.$

Check independence:

$$\begin{aligned} W(y_1,\, y_2) &= \begin{vmatrix} x & x\ln|x|+1 \\ 1 & \ln|x|+1 \end{vmatrix} \\ &= x\ln|x| + x - x\ln|x| - 1 = x - 1 \neq 0 \quad \text{for } x \neq 1. \end{aligned}$$

Hence

$$\{y_1(x),\, y_2(x)\} = \{x,\, x\ln|x| + 1\}$$

is a fundamental set of solutions (on an interval not containing 0 and 1, where $a_2(x) = x^2 - x \neq 0$), and

$$y = c_1 x + c_2\,(x\ln|x| + 1)$$

is the general solution for constants $c_1,\, c_2$.

4.3 Homogeneous Linear Equations with Constant Coefficients

Consider the homogeneous linear equation with **constant coefficients**(常數係數)

$$y'' + ay' + by = 0,$$

where a, b are constants.

Example. For $y'' + y' - 2y = 0$, the coefficients are 1, 1, -2 (all constants).
Try $y = e^x$, then $y' = e^x$ and $y'' = e^x$. Thus

$$e^x + e^x - 2e^x = 0.$$

Hence $y = e^x$ is a solution.

Example. Solve $y'' + 5y' + 6y = 0$.

Sol. Try $y = e^{-x}$, then $y' = -e^{-x}$ and $y'' = e^{-x}$. Thus

$$e^{-x} + 5\left(-e^{-x}\right) + 6e^{-x} = e^{-x}(1 - 5 + 6) = 2e^{-x} \neq 0.$$

So $y = e^{-x}$ is **not** a solution.
Try $y = e^{-2x}$, then $y' = -2e^{-2x}$ and $y'' = 4e^{-2x}$. Substituting(代入), we have

$$4e^{-2x} + 5\left(-2e^{-2x}\right) + 6e^{-2x} = e^{-2x}\left(4 - 10 + 6\right) = 0.$$

So $y = e^{-2x}$ is a solution.
In general, try $y = e^{ax}$, then $y' = ae^{ax}$ and $y'' = a^2e^{ax}$. Thus

$$a^2e^{ax} + 5ae^{ax} + 6e^{ax} = 0 \implies e^{ax}\left(a^2 + 5a + 6\right) = 0.$$

Since $e^{ax} > 0$,

$$a^2 + 5a + 6 = 0 \implies (a + 2)(a + 3) = 0 \implies a = -2, -3.$$

Hence $y_1 = e^{-2x}$ and $y_2 = e^{-3x}$ are solutions, and

$$y_h = c_1e^{-2x} + c_2e^{-3x},\ c_1,\ c_2 \in \mathbb{R}.$$

• Characteristic Equation

For $y'' + ay' + by = 0$, let $y = e^{\lambda x}$, then

$$y' = \lambda e^{\lambda x}, \quad y'' = \lambda^2 e^{\lambda x}.$$

Thus

$$\lambda^2 e^{\lambda x} + a\lambda e^{\lambda x} + be^{\lambda x} = 0 \implies e^{\lambda x}\left(\lambda^2 + a\lambda + b\right) = 0.$$

Since $e^{\lambda x} > 0$, we obtain

$$\lambda^2 + a\lambda + b = 0.$$

Definition. The equation

$$\lambda^2 + a\lambda + b = 0$$

is called the **characteristic equation**(特徵方程式) or **auxiliary equation**(輔助方程式) of $y'' + ay' + by = 0$. Its roots λ_1, λ_2 are called the **eigenvalues**(特徵值).

By the quadratic formula,

$$\lambda_1 = \frac{-a + \sqrt{a^2 - 4b}}{2}, \quad \lambda_2 = \frac{-a - \sqrt{a^2 - 4b}}{2}.$$

There are 3 cases:

Case 1: $a^2 - 4b > 0$, $\lambda_1 \neq \lambda_2$ (distinct real roots). Then

$$y_1 = e^{\lambda_1 x}, \quad y_2 = e^{\lambda_2 x},$$

and

$$y_h = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}.$$

Case 2: $a^2 - 4b = 0$, $\lambda_1 = \lambda_2 = \lambda = -\frac{a}{2}$ (double roots, 重根).
There is only one solution $y_1 = e^{\lambda x}$. To get y_2, we use reduction of order (Section 4.2).
Let $y_2 = u(x)e^{\lambda x}$, then

$$\begin{cases} y_2' = (u' + \lambda u)\, e^{\lambda x} \\ y_2'' = (u'' + 2\lambda u' + \lambda^2 u)\, e^{\lambda x} \end{cases}$$

Substituting into the equation,

$$e^{\lambda x}\left[u'' + (2\lambda + a)u' + \left(\lambda^2 + a\lambda + b\right) u\right] = 0.$$

Since $\lambda = -\frac{a}{2}$, we have $2\lambda + a = 0$; and λ is a root, so $\lambda^2 + a\lambda + b = 0$. Thus

$$u'' = 0 \implies u = x \quad \text{(take the simplest one)}.$$

Hence $y_2 = xe^{\lambda x}$, and

$$y_h = c_1 e^{\lambda x} + c_2 x e^{\lambda x}.$$

Case 3: $a^2 - 4b < 0$, complex conjugate roots(共軛複根).
Since

$$\lambda_{1,2} = \frac{-a \pm \sqrt{a^2 - 4b}}{2} = -\frac{a}{2} \pm \sqrt{\frac{a^2}{4} - b}$$

and $a^2 - 4b < 0 \implies \dfrac{a^2}{4} - b < 0$, let

$$\omega^2 = b - \frac{1}{4}a^2 > 0.$$

Then

$$\lambda_1 = -\frac{a}{2} + i\omega, \quad \lambda_2 = -\frac{a}{2} - i\omega,$$

and

$$y_1 = e^{\left(-\frac{a}{2}+i\omega\right)x}, \quad y_2 = e^{\left(-\frac{a}{2}-i\omega\right)x}, \quad y_h = c_1 y_1 + c_2 y_2.$$

But y_1, y_2 are complex-valued. Recall **Euler's formula**(尤拉等式):

$$\begin{cases} e^{i\theta} = \cos\theta + i\sin\theta \\ e^{-i\theta} = \cos(-\theta) + i\sin(-\theta) = \cos\theta - i\sin\theta \end{cases}$$

Let $\theta = \omega x$, then

$$\begin{aligned} y_1 &= e^{-\frac{a}{2}x} \cdot e^{i\omega x} = e^{-\frac{a}{2}x}(\cos\omega x + i\sin\omega x) \\ y_2 &= e^{-\frac{a}{2}x} \cdot e^{-i\omega x} = e^{-\frac{a}{2}x}(\cos\omega x - i\sin\omega x) \end{aligned}$$

By the superposition principle, any linear combination of y_1, y_2 is also a solution. So we take new solutions:

$$\begin{aligned} y_3 &= \frac{1}{2}y_1 + \frac{1}{2}y_2 = \frac{1}{2}e^{-\frac{a}{2}x} \cdot 2\cos\omega x = e^{-\frac{a}{2}x}\cos\omega x, \\ y_4 &= \frac{1}{2i}y_1 - \frac{1}{2i}y_2 = \frac{1}{2i}e^{-\frac{a}{2}x} \cdot 2i\sin\omega x = e^{-\frac{a}{2}x}\sin\omega x. \end{aligned}$$

y_3, y_4 are real-valued and linearly independent ($W(y_3, y_4) = \omega e^{-ax} \neq 0$). Hence

$$y_h = e^{-\frac{a}{2}x}(A\cos\omega x + B\sin\omega x),$$

where A, B are arbitrary constants(任意常數).

Remark (Summary)**.** For $y''+ay'+by = 0$ with characteristic equation $\lambda^2+a\lambda+b = 0$:

Case	Roots	y_h
$a^2 - 4b > 0$	$\lambda_1 \neq \lambda_2$ (real)	$c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$
$a^2 - 4b = 0$	$\lambda_1 = \lambda_2 = \lambda$	$c_1 e^{\lambda x} + c_2 x e^{\lambda x}$
$a^2 - 4b < 0$	$-\frac{a}{2} \pm i\omega$	$e^{-\frac{a}{2}x}(c_1\cos\omega x + c_2\sin\omega x)$

To solve an IVP of this type, we will use the following steps:

(i) Write the characteristic equation.

(ii) Find λ_1, λ_2.

(iii) Find $y_1(x)$, $y_2(x)$.

(iv) $y_h(x) = c_1 y_1(x) + c_2 y_2(x)$.

(v) Substitute the I.C. to find c_1, c_2.

Example. Solve $y'' + 0.4y' + 9.04y = 0$ with $y(0) = 0$, $y'(0) = 3$.

Sol. (i) The characteristic equation is

$$\lambda^2 + 0.4\lambda + 9.04 = 0.$$

(ii) Then

$$\lambda = \frac{-0.4 \pm \sqrt{0.16 - 36.16}}{2} = \frac{-0.4 \pm \sqrt{-36}}{2} = -0.2 \pm 3i.$$

So $\lambda_1 = -0.2 + 3i$ and $\lambda_2 = -0.2 - 3i$, which is Case 3 with $\dfrac{a}{2} = 0.2$, $\omega = 3$.

(iii) $y_3(x) = e^{-0.2x}\cos 3x$ and $y_4(x) = e^{-0.2x}\sin 3x$.

(iv) The homogeneous solution is

$$y_h(x) = c_1 e^{-0.2x}\cos 3x + c_2 e^{-0.2x}\sin 3x = e^{-0.2x}\left(c_1\cos 3x + c_2 \sin 3x\right).$$

(v) Compute

$$y_h'(x) = e^{-0.2x}\left[(-0.2c_1 + 3c_2)\cos 3x + (-3c_1 - 0.2c_2)\sin 3x\right].$$

Let $y(0) = 0$ and $y'(0) = 3$, we have

$$\begin{cases} y(0) = c_1 = 0 \\ y'(0) = -0.2c_1 + 3c_2 = 3 \end{cases} \implies c_1 = 0,\ c_2 = 1.$$

Hence

$$y(x) = e^{-0.2x}\sin 3x$$

is the solution of the IVP.

Example. Solve $4y'' + 4y' + 17y = 0$ with $y(0) = -1$, $y'(0) = 2$.

Sol. Dividing by 4,

$$y'' + y' + \frac{17}{4}y = 0.$$

The characteristic equation is

$$\lambda^2 + \lambda + \frac{17}{4} = 0 \implies \lambda = \frac{-1 \pm \sqrt{1 - 17}}{2} = -\frac{1}{2} \pm 2i.$$

This is Case 3 with $\omega = 2$, so

$$y_h(x) = e^{-\frac{1}{2}x}\left(c_1 \cos 2x + c_2 \sin 2x\right).$$

Then

$$y_h'(x) = e^{-\frac{1}{2}x}\left[\left(-\frac{1}{2}c_1 + 2c_2\right)\cos 2x + \left(-2c_1 - \frac{1}{2}c_2\right)\sin 2x\right].$$

Let $y(0) = -1$ and $y'(0) = 2$,

$$\begin{cases} c_1 = -1 \\ -\dfrac{1}{2}c_1 + 2c_2 = 2 \end{cases} \implies \frac{1}{2} + 2c_2 = 2 \implies c_2 = \frac{3}{4}.$$

Hence

$$y(x) = -e^{-\frac{1}{2}x}\cos 2x + \frac{3}{4}e^{-\frac{1}{2}x}\sin 2x.$$

• Higher-Order Equations

The same method works for higher-order equations: let $y = e^{\lambda x}$ and solve the characteristic equation.

Example. Solve $y''' + 3y'' - 4y = 0$.

Sol. (i) The characteristic equation is

$$\lambda^3 + 3\lambda^2 + 0 \cdot \lambda - 4 = 0 \implies \lambda^3 + 3\lambda^2 - 4 = 0.$$

(ii) Since $\lambda = 1$ satisfies $1 + 3 - 4 = 0$, $(\lambda - 1)$ is a factor. By polynomial division,

$$\lambda^3 + 3\lambda^2 - 4 = (\lambda - 1)\left(\lambda^2 + 4\lambda + 4\right) = (\lambda - 1)(\lambda + 2)^2.$$

Thus $\lambda_1 = 1$ and $\lambda_2 = \lambda_3 = -2$ (double root).

(iii) By Case 1 and Case 2,

$$y_1 = e^x, \quad y_2 = e^{-2x}, \quad y_3 = xe^{-2x}.$$

Hence

$$y_h(x) = c_1 e^x + c_2 e^{-2x} + c_3 x e^{-2x}.$$